Inset-Fed Microstrip Patch Antenna Design and CST Validation

Analytical Synthesis, Impedance Matching, and Radiation Verification

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Independent Engineering Portfolio Project

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Portfolio Context and Evidence

Project Context. This work originated from a graduate-level antenna engineering simulation project. The present document is an independently prepared public portfolio edition intended to demonstrate rectangular microstrip-patch synthesis, inset-feed matching, full-wave electromagnetic simulation, radiation analysis, engineering interpretation, and technical documentation.

Public Portfolio Edition. The original academic submission has been reorganized for professional and educational use. Student-identification information, course-specific grading strategy, rubric language, institutional branding, and administrative submission material have been removed. The 915 MHz design frequency and inset-feed topology are retained as the engineering specification of the public study. This document does not represent an official publication or endorsement by any university, instructor, employer, software vendor, or material supplier.

Evidence Classification. Analytical evidence includes the transmission-line-model estimates for patch width, effective dielectric constant, fringing-field extension, physical patch length, and inset depth, together with an independently executable Python verification script. Simulated evidence consists of preserved CST Studio Suite Learning Edition screenshots for the final geometry, reflection coefficient, electric and magnetic fields, and far-field patterns. No antenna prototype was fabricated and no calibrated VNA or antenna-range measurements were conducted. All reported antenna performance is therefore analytical or simulated, not measured.

Simulation Scope. The public evidence set contains exported screenshots and LaTeX source but not the native CST project or raw numerical curve exports. The exact resonance and minimum S_{11} are supported by a displayed marker. Far-field quantities are read from the preserved CST plots at 0.915 GHz. Formal mesh-convergence and boundary-distance convergence studies were not preserved. The patch and ground were modeled as PEC, so conductor loss is not represented; dielectric loss was retained through the substrate loss tangent. These qualifications constrain the interpretation of the results.

Prepared as part of the Hossein Molhem Engineering Portfolio.

Abstract

This report presents the analytical design and CST full-wave simulation of an inset-fed rectangular microstrip patch antenna operating near 915 MHz. The antenna uses a Rogers RT/duroid 5880 substrate with relative permittivity $\varepsilon_r = 2.20$, loss tangent $\tan\delta = 0.0009$, and thickness $h = 1.575$ mm. Standard transmission-line-model relations give an initial patch width of 129.51 mm, effective dielectric constant of 2.1605, fringing extension of 0.834 mm, and physical patch length of 109.79 mm. The documented 50-Ω feed-line width is 4.89 mm, and the cosine-squared input-resistance estimate gives an inset depth of 38.13 mm for the final patch length.

The final CST geometry uses $W = 129.5$ mm, $L = 108.2$ mm, a 4.89 mm feed line, a 6.89 mm inset-notch width, and a 38.2 mm inset depth. The preserved CST marker places the matching minimum at 0.9186 GHz with $S_{11,\mathrm{min}} = -22.14379$ dB. This corresponds to a positive return loss of 22.14379 dB and a frequency deviation of 3.6 MHz, or 0.393%, from the 915 MHz target.

At 0.915 GHz, the CST far-field plots display a peak directivity of 7.01 dBi and a broadside main beam. The $\phi = 0°$ cut shows a main-beam direction of 1°, a 3-dB angular width of 82.3°, and a displayed sidelobe level of −19.2 dB. The $\phi = 90°$ cut shows a main-beam direction of 0° and a 3-dB angular width of 90.8°. The study connects closed-form synthesis, parameterized model construction, resonance adjustment, impedance matching, field visualization, and radiation verification.

All performance results are analytical or simulated. No fabricated prototype or measured antenna data are claimed. The absence of raw CST exports, native model files, and formal mesh- and boundary-convergence records limits the conclusions to the preserved evidence package.

Project Objective

The objective of this project is to establish a traceable antenna-engineering workflow for an inset-fed rectangular microstrip patch operating near 915 MHz. The work connects first-order transmission-line synthesis to a parameterized CST model and evaluates the final input match, near-field behavior, and broadside radiation characteristics.

The project has four principal technical objectives:

1. Calculate the initial patch width and length from the design frequency, substrate permittivity, substrate thickness, effective dielectric constant, and fringing-field correction.
2. Define a practical inset-fed geometry using a documented 50-Ω microstrip width and a first-order inset-depth estimate based on the patch input resistance.
3. Construct and tune the antenna in CST Studio Suite, then quantify the final resonant frequency and reflection coefficient relative to the configured port.
4. Evaluate the preserved electric-field, magnetic-field, 3D directivity, and principal-plane radiation-pattern evidence at the 915 MHz design frequency.

A secondary objective is to distinguish exact displayed markers, values read directly from CST information panels, analytical calculations, and qualitative field evidence. The report also identifies the numerical and experimental work still required before hardware-performance claims would be justified.

The final outcome is a concise, university-style public engineering report consistent with the other antenna projects in this portfolio. All reported performance is analytical or simulated; no fabricated or measured antenna result is claimed.

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Chapter 1

Inset-Fed Microstrip Patch Antenna Design and CST Validation

1.1 Project Overview

Microstrip patch antennas provide a low-profile and mechanically simple radiating structure for systems that require planar integration. A rectangular patch consists of a conducting plate above a ground plane, separated by a dielectric substrate. The patch behaves approximately as a resonant cavity, while the fringing fields at the open edges produce radiation [1].

This study develops an inset-fed rectangular patch near 915 MHz using Rogers RT/-duroid 5880. Analytical transmission-line relations establish the starting geometry, and CST Studio Suite is used to refine the physical length and feed position. The final evidence set includes geometry, S_{11}, electric-field, magnetic-field, and far-field plots.

Table 1.1: Principal design and simulation results.

Quantity	Result
Design frequency	0.915 GHz
Substrate	Rogers RT/duroid 5880, $\varepsilon_r = 2.20$, $h = 1.575$ mm
Analytical patch dimensions	$W = 129.51$ mm, $L = 109.79$ mm
Final CST patch dimensions	$W = 129.5$ mm, $L = 108.2$ mm
Final inset depth	38.2 mm
CST matching minimum	0.9186 GHz, $S_{11} = -22.14379$ dB
Frequency error	3.6 MHz, or 0.393%
Displayed peak directivity at 0.915 GHz	7.01 dBi
Displayed 3-dB beamwidths	82.3° ($\phi = 0°$), 90.8° ($\phi = 90°$)
Evidence status	Analytical and simulated; no measured validation

1.2 Engineering Scope and Evidence

The public project is defined by the engineering specification in Table 1.2. The original administrative mechanism used to select the frequency and feed type is intentionally omitted; the resulting 915 MHz inset-fed configuration is retained as the fixed design input.

Table 1.2: Public design specification.

Category	Parameter	Specified value
Electrical	Design frequency	915 MHz
Electrical	Feed topology	Inset-fed microstrip line
Material	Substrate	Rogers RT/duroid 5880
Material	Relative permittivity	2.20
Material	Loss tangent	0.0009
Geometry	Substrate thickness	1.575 mm
Geometry	Metal thickness used geometrically	0.035 mm
Simulation	Frequency range	0.6–1.2 GHz
Simulation	Boundaries	Open (add space), all sides
Simulation	Monitors	E field, H field, and far field at 0.915 GHz

1.2.1 Evidence Classes

Analytical evidence consists of the equations and independently executable script used to regenerate the initial patch dimensions and inset-depth estimate.

Simulated evidence consists of CST screenshots. The exact S_{11} minimum is supported by a marker in the preserved plot. Directivity, main-beam direction, beamwidth, and displayed efficiency values are read from the CST far-field information panels.

Measured evidence is absent. No fabricated antenna, VNA sweep, calibrated gain measurement, radiation-pattern measurement, or material characterization is included.

1.2.2 Public Claim Boundary

The report claims only that the documented analytical model and preserved CST evidence support the stated simulated behavior. It does not claim manufacturing readiness, broadband performance, tolerance robustness, regulatory compliance, or measured hardware performance.

1.3 Theory and Analytical Design

The initial geometry follows the conventional transmission-line approximation for the dominant rectangular-patch resonance [1]. The free-space wavelength at the design frequency is

$$\lambda_0 = \frac{c}{f_0} = 327.64 \text{ mm}. \tag{1.1}$$

1.3.1 Patch Width

The initial width is

$$W = \frac{c}{2f_0}\sqrt{\frac{2}{\varepsilon_r + 1}}, \tag{1.2}$$

which gives

$$W = 129.51 \text{ mm}. \tag{1.3}$$

1.3.2 Effective Permittivity and Fringing Extension

The effective dielectric constant is estimated as

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + 12\frac{h}{W}\right)^{-1/2}, \tag{1.4}$$

which yields $\varepsilon_{\text{eff}} = 2.1605$.

The fringing-field extension is

$$\Delta L = 0.412h \frac{(\varepsilon_{\text{eff}} + 0.3)(W/h + 0.264)}{(\varepsilon_{\text{eff}} - 0.258)(W/h + 0.8)}, \tag{1.5}$$

producing $\Delta L = 0.834$ mm.

1.3.3 Effective and Physical Length

The effective resonant length is

$$L_{\text{eff}} = \frac{c}{2f_0\sqrt{\varepsilon_{\text{eff}}}} = 111.45 \text{ mm}, \tag{1.6}$$

and the physical starting length is

$$L = L_{\text{eff}} - 2\Delta L = 109.79 \text{ mm}. \tag{1.7}$$

1.3.4 Feed Width and Inset Depth

The documented design uses a 4.89 mm microstrip feed width as the nominal 50-Ω value for the selected substrate. The inset depth is estimated from

$$R_{\text{in}}(y_0) = R_{\text{edge}} \cos^2\left(\frac{\pi y_0}{L}\right). \tag{1.8}$$

Using $R_{\text{edge}} = 250\ \Omega$, a target resistance of 50 Ω, and the final patch length $L = 108.2$ mm gives

$$y_0 = \frac{L}{\pi} \cos^{-1}\sqrt{\frac{50}{250}} = 38.13 \text{ mm}. \tag{1.9}$$

The final CST inset depth is 38.2 mm.

Table 1.3: Analytical starting values and final CST dimensions.

Quantity	Symbol	Analytical value	Final CST value
Patch width	W	129.51 mm	129.5 mm
Patch length	L	109.79 mm	108.2 mm
Effective dielectric constant	ε_{eff}	2.1605	–
Fringing extension	ΔL	0.834 mm	–
Feed width	W_f	Documented 4.89 mm	4.89 mm
Inset depth	d_{inset}	38.13 mm	38.2 mm
Inset-notch width	W_{notch}	6.89 mm	6.89 mm

1.4 CST Model and Simulation Method

The model was created in CST Studio Suite Learning Edition using the planar-antenna workflow. Geometry was defined in millimeters and frequency in GHz. The simulation range of 0.6–1.2 GHz brackets both the 0.915 GHz design point and the final matching minimum.

1.4.1 Material and Conductor Models

Rogers RT/duroid 5880 was represented by $\varepsilon_r = 2.20$ and $\tan\delta = 0.0009$, values documented by Rogers for the laminate [2], [3]. The patch and ground conductors were modeled as PEC. Their 0.035 mm thickness is therefore geometrical; finite copper conductivity and conductor loss are not included.

1.4.2 Final Geometry

The patch was assembled from a main body and two inset arms, while a central feed strip extended to the waveguide port. Figure 1.1 shows the preserved CST model.

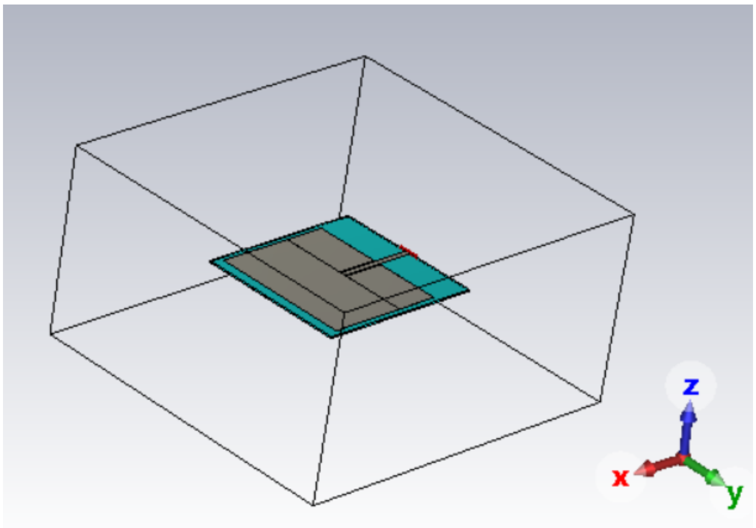

Figure 1.1: Preserved CST geometry of the final inset-fed rectangular microstrip patch antenna.

Table 1.4: Final CST object coordinates. All dimensions are in millimeters.

Object	x_{min}	x_{max}	y_{min}	y_{max}	z_{min}	z_{max}
Substrate	-35.0	118.2	-69.75	69.75	0.000	1.575
Ground	-35.0	118.2	-69.75	69.75	-0.035	0.000
Patch body	38.2	108.2	-64.75	64.75	1.575	1.610
Upper inset arm	0.0	38.2	3.445	64.75	1.575	1.610
Lower inset arm	0.0	38.2	-64.75	-3.445	1.575	1.610
Feed line	-35.0	38.2	-2.445	2.445	1.575	1.610

1.4.3 Port, Boundaries, and Monitors

A waveguide port was placed at the left substrate edge and oriented in the positive x direction. Open-add-space boundaries were used on all sides. E-field, H-field, and far-field monitors were defined at 0.915 GHz.

Table 1.5: Documented port and solver configuration.

Setting	**Documented value**
Excitation	Waveguide port 1
Port plane	$x = -35$ mm
Port window	$y = -10$ to 10 mm; $z = -0.035$ to 4 mm
Propagation direction	Positive x
Frequency range	0.6–1.2 GHz
Boundary condition	Open (add space), all sides
Field monitors	E field, H field, far field at 0.915 GHz

1.5 Simulation Results and Verification

1.5.1 Reflection Coefficient and Resonance

Figure 1.2 contains the exact displayed CST marker

$$f_{\mathrm{res}} = 0.9186 \text{ GHz}, \qquad S_{11,\mathrm{min}} = -22.14379 \text{ dB}. \tag{1.10}$$

The corresponding positive return loss is

$$\mathrm{RL} = -S_{11,\mathrm{dB}} = 22.14379 \text{ dB}. \tag{1.11}$$

This terminology is important: the plotted quantity is the negative dB magnitude of S_{11}, while return loss is conventionally reported as a positive number.

The frequency error is

$$\frac{0.9186 - 0.915}{0.915} \times 100 = 0.393\%. \tag{1.12}$$

The marker also implies $|\Gamma| = 0.0782$ and a derived VSWR of approximately 1.17, relative to the configured CST port reference.

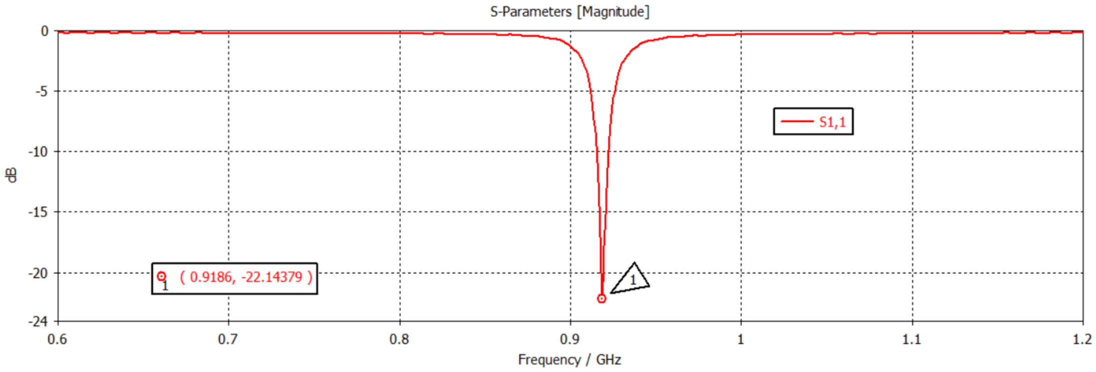

Figure 1.2: Preserved CST reflection-coefficient plot. The marker identifies the matching minimum at 0.9186 GHz and $S_{11} = -22.14379$ dB.

Table 1.6: Final input-match results and derived quantities.

Quantity	Value	Evidence type
Design frequency	0.915 GHz	Specification
Matching-minimum frequency	0.9186 GHz	Exact displayed marker
Minimum S_{11}	-22.14379 dB	Exact displayed marker
Positive return loss	22.14379 dB	Derived from marker
Reflection magnitude $\lvert\Gamma\rvert$	0.0782	Derived from marker
VSWR	1.17	Derived from marker
Frequency error	0.393%	Derived from marker

1.5.2 Electric- and Magnetic-Field Evidence

The preserved field monitors were evaluated at 0.915 GHz. Figures 1.3 and 1.4 show concentration around the patch, inset, and feed region. These images support resonant excitation qualitatively, but no raw field export is available for independent quantitative post-processing.

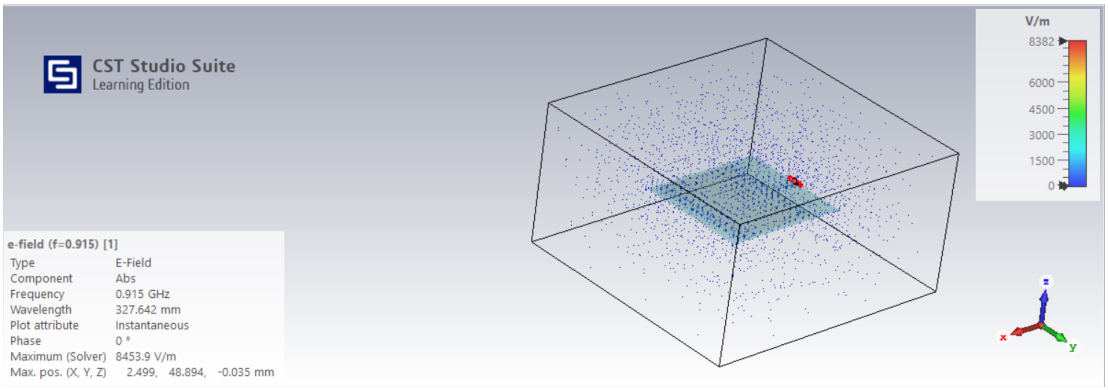

Figure 1.3: Simulated electric-field distribution at 0.915 GHz.

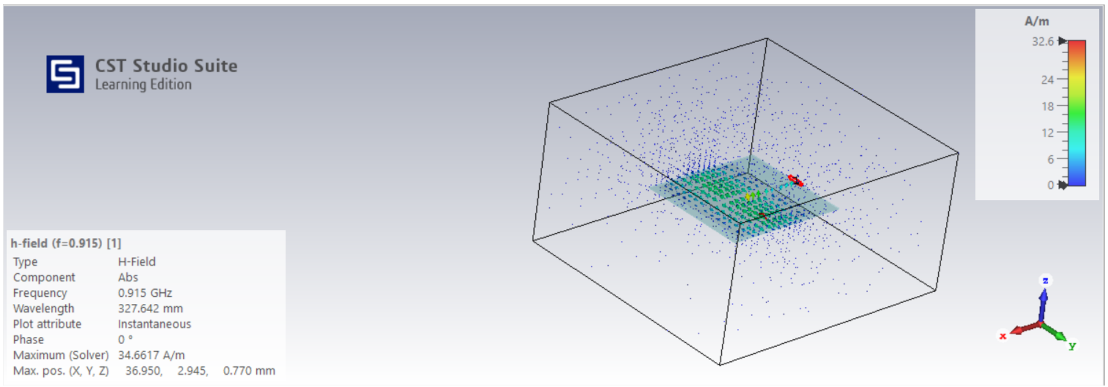

Figure 1.4: Simulated magnetic-field distribution at 0.915 GHz.

1.5.3 Three-Dimensional Far Field

The CST 3D plot in Fig. 1.5 displays a broadside main lobe and a peak directivity of 7.01 dBi at 0.915 GHz. The information panel also displays radiation efficiency of 0.03436 dB and total efficiency of -0.4787 dB. Because the radiation efficiency is slightly positive in dB, it corresponds to a value marginally above 100% and is treated as a numerical-display anomaly rather than a validated efficiency claim. The total-efficiency display is retained as supporting evidence but is not used as a principal performance KPI without convergence and raw data.

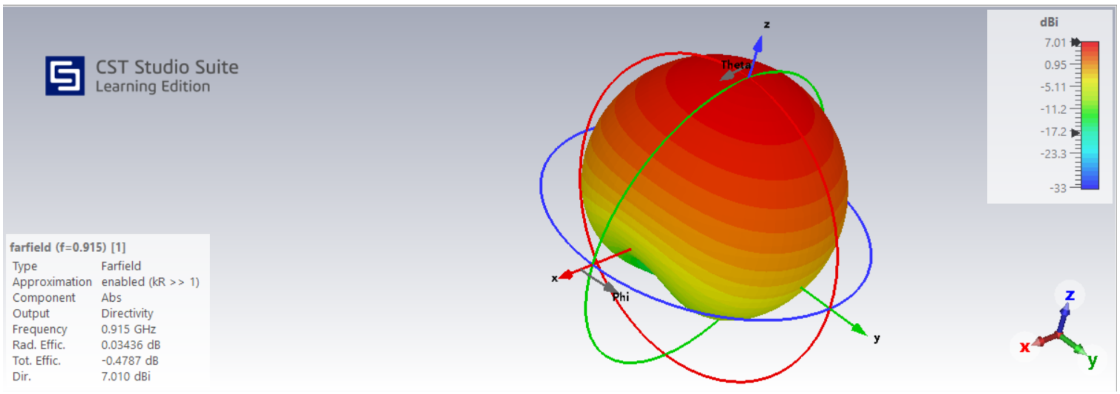

Figure 1.5: CST 3D far-field directivity at 0.915 GHz. The displayed peak is 7.01 dBi.

1.5.4 Principal-Plane Radiation Patterns

Figures 1.6 and 1.7 show the two documented principal-plane cuts. Both support broadside radiation. The $\phi = 0°$ cut includes a displayed sidelobe level of -19.2 dB; no corresponding sidelobe value is shown in the $\phi = 90°$ screenshot.

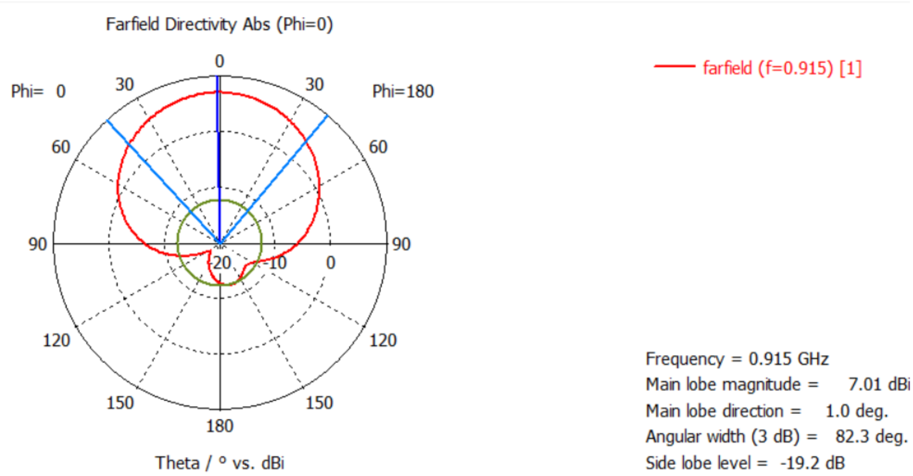

Figure 1.6: CST principal-plane directivity pattern at 0.915 GHz for $\phi = 0°$.

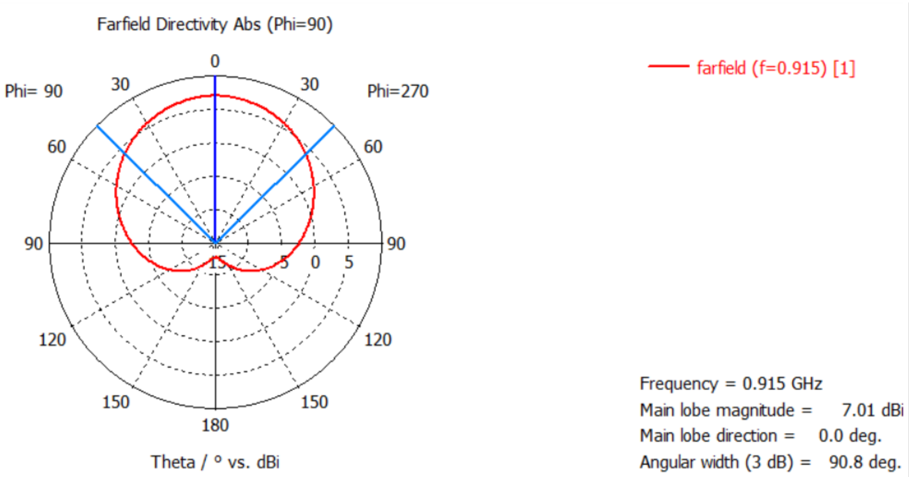

Figure 1.7: CST principal-plane directivity pattern at 0.915 GHz for $\phi = 90°$.

Table 1.7: Displayed principal-plane far-field quantities at 0.915 GHz.

Cut	Peak	Direction	3-dB width	Displayed SLL
$\phi = 0°$	7.01 dBi	1.0°	82.3°	-19.2 dB
$\phi = 90°$	7.01 dBi	0.0°	90.8°	Not displayed

1.6 Engineering Discussion and Numerical Reliability

1.6.1 Analytical-to-Full-Wave Adjustment

The final patch width is effectively unchanged from the analytical starting value, while the final patch length is 1.59 mm shorter than the analytical estimate. This is a 1.45% reduction. Such adjustment is consistent with the role of the analytical model as a first-order starting point: the full-wave solution incorporates the actual inset, feed transition, finite substrate and ground, port geometry, and numerical boundary configuration.

1.6.2 Impedance Matching

The inset location controls the resistance sampled along the patch. The analytical 38.13 mm estimate and the final 38.2 mm CST value agree closely, indicating that the cosine-squared resistance model provided a useful initial feed location. The final S_{11}

marker demonstrates a well-defined simulated match near the design frequency. It does not establish tolerance robustness or measured matching after fabrication.

1.6.3 Radiation Behavior

The 3D and principal-plane plots show a broadside pattern consistent with the expected radiation from the fringing fields at the patch edges. The two principal planes have different 3-dB widths, which is expected because the rectangular aperture dimensions and field distributions are not identical in the two planes.

1.6.4 Numerical Qualifications

The following limitations materially affect interpretation:

1. The public archive does not contain the native CST model, raw S-parameter export, raw far-field data, mesh statistics, or solver log.
2. No mesh-convergence or boundary-distance convergence study is preserved. The final resonance and far-field values therefore represent the documented model, not a formally convergence-qualified solution.
3. The field and far-field monitors are at 0.915 GHz, whereas the matching minimum occurs at 0.9186 GHz. The two frequencies differ by only 0.393%, but the distinction is retained explicitly.
4. PEC metallization omits conductor loss. The reported directivity is not gain, and no finite-conductivity sensitivity study is available.
5. The slightly positive displayed radiation efficiency in dB is not physically admissible as an exact efficiency result and is treated as a small numerical-display artifact. Efficiency values are therefore not promoted as verified KPIs.
6. No fabrication, connector transition, manufacturing tolerance, material tolerance, VNA calibration, or antenna-range validation is included.

These limitations do not invalidate the documented design workflow, resonance result, or qualitative broadside pattern. They define the boundary between a successful simulation study and a measurement-qualified antenna design.

1.7 Engineering Verification Matrix

The matrix in Table 1.8 links each principal public claim to its evidence source and qualification.

Table 1.8: Engineering verification matrix.

ID	Public claim	Evidence class	Evidence location	Qualification
V1	The 915 MHz analytical patch width is 129.51 mm and physical length is 109.79 mm.	Analytical	Eqs. (1.2)–(1.7); Appendix A	First-order transmission-line approximation.
V2	The final documented CST geometry uses $W = 129.5$ mm, $L = 108.2$ mm, and a 38.2 mm inset.	Simulated model record	Table 1.4; Fig. 1.1	Native CST model is not in the public archive.
V3	The final matching minimum is 0.9186 GHz with $S_{11} = -22.14379$ dB.	Exact simulated marker	Fig. 1.2; Eq. (1.10)	Relative to the configured CST port; raw export unavailable.
V4	The frequency deviation from 915 MHz is 0.393%.	Derived analytical result	Eq. (1.12); verification script	Uses the exact displayed frequency marker.
V5	The far-field plot displays a 7.01 dBi broadside peak at 0.915 GHz.	Displayed simulation result	Fig. 1.5; Table 1.7	Directivity, not realized gain; no raw far-field export.
V6	The displayed 3-dB widths are 82.3° and 90.8° in the two principal planes.	Displayed simulation result	Figs. 1.6 and 1.7	Values read from CST information panels.
V7	The E- and H-field images support resonant excitation around the patch and feed.	Qualitative simulation evidence	Figs. 1.3 and 1.4	No raw field data for independent quantitative analysis.
V8	The conductor model does not include copper loss.	Model qualification	Section 1.4	Patch and ground are PEC despite finite geometric thickness.

Continued on next page

ID	Public claim	Evidence class	Evidence location	Qualification
V9	Efficiency is not promoted as a verified KPI.	Numerical qualification	Section 1.6; Fig. 1.5	Displayed radiation efficiency is slightly above 0 dB.
V10	No measured hardware performance is claimed.	No measurement	Portfolio Context and Evidence	Requires fabrication, VNA, and radiation-pattern testing.

1.8 Conclusion

An inset-fed rectangular microstrip patch antenna was analytically synthesized and modeled in CST Studio Suite for operation near 915 MHz. The first-order design produced $W = 129.51$ mm and $L = 109.79$ mm on Rogers RT/duroid 5880. The final CST geometry used $W = 129.5$ mm, $L = 108.2$ mm, a 4.89 mm feed line, and a 38.2 mm inset depth.

The exact displayed CST marker placed the matching minimum at 0.9186 GHz with $S_{11} = -22.14379$ dB, equivalent to a positive return loss of 22.14379 dB. The frequency deviation from the 915 MHz target is 0.393%. At the design frequency, the preserved far-field evidence shows a 7.01 dBi peak directivity and broadside radiation, with displayed 3-dB widths of 82.3° and 90.8° in the two principal planes.

The project demonstrates a coherent workflow from analytical patch synthesis through inset-feed selection, CST construction, input-match evaluation, field visualization, and far-field interpretation. The strongest conclusions are the traceable analytical dimensions, exact displayed S_{11} marker, and documented broadside directivity patterns. Formal mesh and boundary convergence, finite-conductivity modeling, raw-data export, fabrication, VNA testing, and radiation-range measurements remain future work.

Appendix A

Analytical Verification

This appendix provides the executable Python script used to regenerate the principal analytical dimensions and derived input-match quantities.

A.1 Verification Script

```python
"""Analytical verification for the 915 MHz inset-fed microstrip patch portfolio
    report."""

import math

C0 = 299_792_458.0
F0_HZ = 915e6
EPS_R = 2.20
H_M = 1.575e-3
TARGET_Z_OHM = 50.0
EDGE_RESISTANCE_OHM = 250.0

FINAL_LENGTH_MM = 108.2
FINAL_WIDTH_MM = 129.5
DOCUMENTED_FEED_WIDTH_MM = 4.89
NOTCH_GAP_MM = 1.0
FINAL_RESONANCE_GHZ = 0.9186
FINAL_S11_DB = -22.14379

lambda0_m = C0 / F0_HZ
width_m = C0 / (2.0 * F0_HZ) * math.sqrt(2.0 / (EPS_R + 1.0))
eps_eff = (
    (EPS_R + 1.0) / 2.0
    + (EPS_R - 1.0) / 2.0 * (1.0 + 12.0 * H_M / width_m) ** -0.5
)
delta_l_m = (
    0.412
```

```
    * H_M
    * ((eps_eff + 0.3) * (width_m / H_M + 0.264))
    / ((eps_eff - 0.258) * (width_m / H_M + 0.8))
)
effective_length_m = C0 / (2.0 * F0_HZ * math.sqrt(eps_eff))
physical_length_m = effective_length_m - 2.0 * delta_l_m

inset_depth_mm = (
    FINAL_LENGTH_MM
    / math.pi
    * math.acos(math.sqrt(TARGET_Z_OHM / EDGE_RESISTANCE_OHM))
)
notch_width_mm = DOCUMENTED_FEED_WIDTH_MM + 2.0 * NOTCH_GAP_MM
frequency_error_mhz = FINAL_RESONANCE_GHZ * 1000.0 - F0_HZ / 1e6
frequency_error_percent = frequency_error_mhz / (F0_HZ / 1e6) * 100.0
return_loss_db = -FINAL_S11_DB
gamma_mag = 10.0 ** (FINAL_S11_DB / 20.0)
vswr = (1.0 + gamma_mag) / (1.0 - gamma_mag)
length_adjustment_mm = FINAL_LENGTH_MM - physical_length_m * 1e3
length_adjustment_percent = length_adjustment_mm / (physical_length_m * 1e3) * 100.0

print("Inset-Fed Microstrip Patch Analytical Verification")
print("=" * 72)
print(f"Design frequency: {F0_HZ / 1e9:.3f} GHz")
print(f"Relative permittivity: {EPS_R:.2f}")
print(f"Substrate thickness: {H_M * 1e3:.3f} mm")
print()
print("Analytical starting dimensions")
print(f"Free-space wavelength: {lambda0_m * 1e3:.2f} mm")
print(f"Patch width: {width_m * 1e3:.2f} mm")
print(f"Effective dielectric constant: {eps_eff:.4f}")
print(f"Fringing extension: {delta_l_m * 1e3:.3f} mm")
print(f"Effective patch length: {effective_length_m * 1e3:.2f} mm")
print(f"Physical patch length: {physical_length_m * 1e3:.2f} mm")
print()
print("Final documented geometry")
print(f"Patch width: {FINAL_WIDTH_MM:.1f} mm")
print(f"Patch length: {FINAL_LENGTH_MM:.1f} mm")
print(f"Documented feed width: {DOCUMENTED_FEED_WIDTH_MM:.2f} mm")
print(f"Inset-notch width: {notch_width_mm:.2f} mm")
print(f"Analytical inset-depth estimate: {inset_depth_mm:.2f} mm")
print(f"Length adjustment from analytical value: {length_adjustment_mm:.2f} mm
    ({length_adjustment_percent:.2f}%)")
print()
print("Final CST marker and derived quantities")
print(f"Matching-minimum frequency: {FINAL_RESONANCE_GHZ:.4f} GHz")
print(f"Minimum S11: {FINAL_S11_DB:.5f} dB")
print(f"Positive return loss: {return_loss_db:.5f} dB")
print(f"Reflection magnitude: {gamma_mag:.4f}")
print(f"VSWR: {vswr:.3f}")
print(f"Frequency error: {frequency_error_mhz:.1f} MHz
    ({frequency_error_percent:.3f}%)")
```

Listing A.1: Python verification of the patch dimensions, inset estimate, and derived matching quantities.

A.2 Console Output

```
Inset-Fed Microstrip Patch Analytical Verification
========================================================================
Design frequency: 0.915 GHz
Relative permittivity: 2.20
Substrate thickness: 1.575 mm

Analytical starting dimensions
Free-space wavelength: 327.64 mm
Patch width: 129.51 mm
Effective dielectric constant: 2.1605
Fringing extension: 0.834 mm
Effective patch length: 111.45 mm
Physical patch length: 109.79 mm

Final documented geometry
Patch width: 129.5 mm
Patch length: 108.2 mm
Documented feed width: 4.89 mm
Inset-notch width: 6.89 mm
Analytical inset-depth estimate: 38.13 mm
Length adjustment from analytical value: -1.59 mm (-1.44%)

Final CST marker and derived quantities
Matching-minimum frequency: 0.9186 GHz
Minimum S11: -22.14379 dB
Positive return loss: 22.14379 dB
Reflection magnitude: 0.0781
VSWR: 1.170
Frequency error: 3.6 MHz (0.393%)
```

Listing A.2: Captured output from the analytical verification script.

Appendix B

Simulation Evidence Record

B.1 Run Manifest

Table B.1: Preserved final CST evidence record.

Record	Frequency or value	Evidence
Final geometry	$W = 129.5$ mm, $L = 108.2$ mm	Geometry screenshot and source tables
Input match	0.9186 GHz, -22.14379 dB	Exact S_{11} marker
E-field monitor	0.915 GHz	CST screenshot
H-field monitor	0.915 GHz	CST screenshot
3D far field	0.915 GHz, 7.01 dBi	CST information panel
$\phi = 0°$ cut	82.3° 3-dB width	CST information panel
$\phi = 90°$ cut	90.8° 3-dB width	CST information panel

B.2 Evidence Availability

The public package includes all screenshots used in the report, the LaTeX source, analytical verification code, and machine-readable key-result summaries. The native CST project, raw curve exports, and solver/mesh logs are not included in the supplied source archive and are therefore not claimed as reproducibility assets.

References

[1] W. L. Stutzman and G. A. Thiele, *Antenna Theory and Design*, 3rd ed. Hoboken, NJ: John Wiley & Sons, 2013.

[2] Rogers Corporation. "RT/duroid® 5880 Laminates," Rogers Corporation, Accessed: Jul. 6, 2026. [Online]. Available: https://www.rogerscorp.com/advanced-electronics-solutions/rt-duroid-laminates/rt-duroid-5880-laminates.

[3] Rogers Corporation. "RT/duroid® 5870/5880 High Frequency Laminates Data Sheet," Rogers Corporation, Accessed: Jul. 6, 2026. [Online]. Available: https://www.rogerscorp.com/-/media/project/rogerscorp/documents/advanced-electronics-solutions/english/data-sheets/rt-duroid-5870---5880-data-sheet.pdf.